

ouees-202606 topic 03

- **Packet switching**
- **Routing basics**

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On the internet

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Lecture notes and reporting

- <https://github.com/jj1bdx/oueees-202606-public/>
- Check out the README.md file and the issues!
- Keyword at the end of the talk
- URL for submitting the report at the end of the talk

Packet Switching

What if you can split a stream into the *packets* and let them be delivered through *different links* for each packet?

How to form a packet (1/2)

- Split a stream into multiple pieces of data (payloads)

ABCDEFHIJ → ABC DEF HIJ

- Put a header on each piece

ABC DEF HIJ → P1-ABC P2-DEF P3-HIJ

How to form a packet (2/2)

- **Add source and destination addresses to each packet**

P1-ABC P2-DEF P3-HIJ

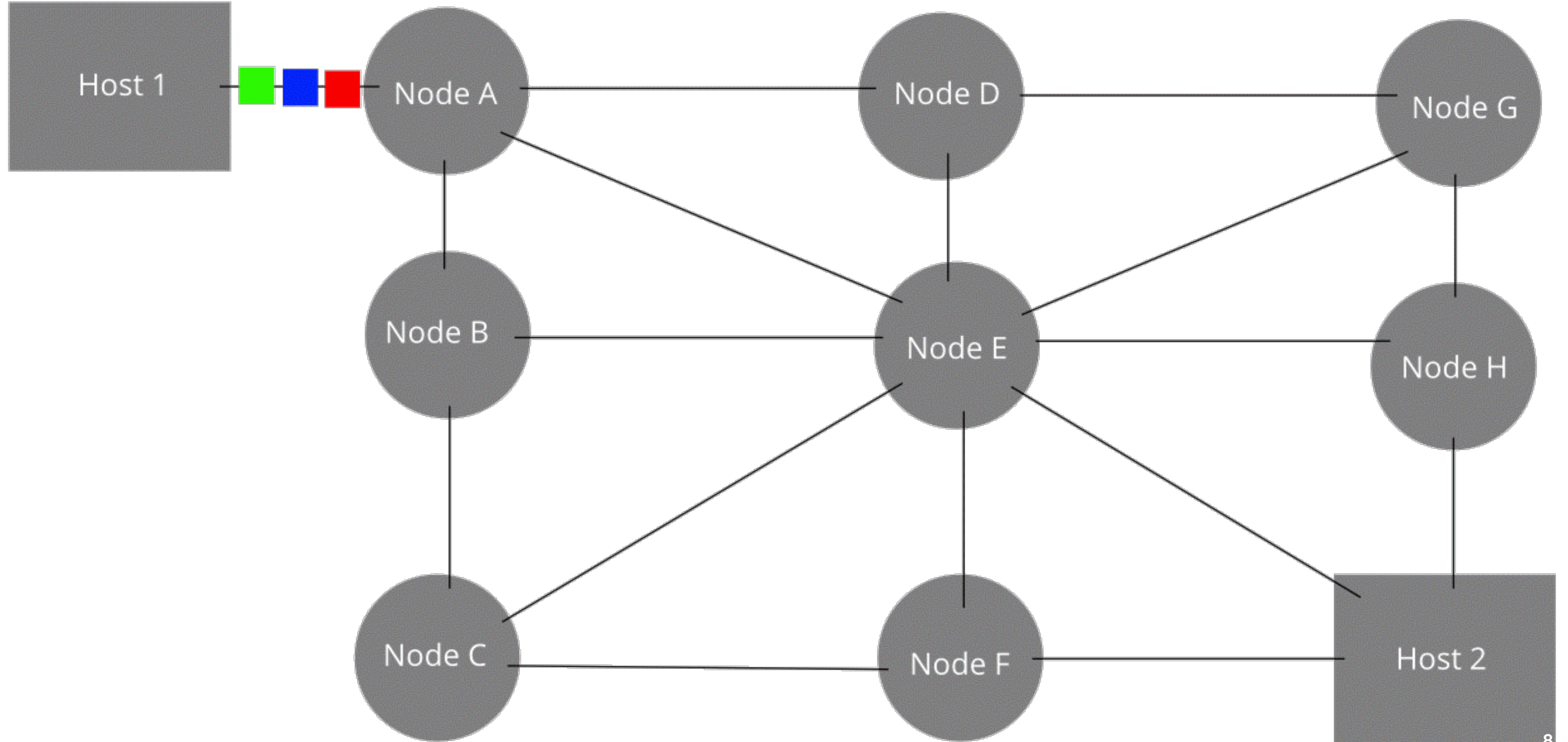
—> FromXtoY-P1-ABC

FromXtoY-P2-DEF

FromXtoY-P3-HIJ

- **Then send them on the network!**

The original message is **Green**, **Blue**, **Red**.

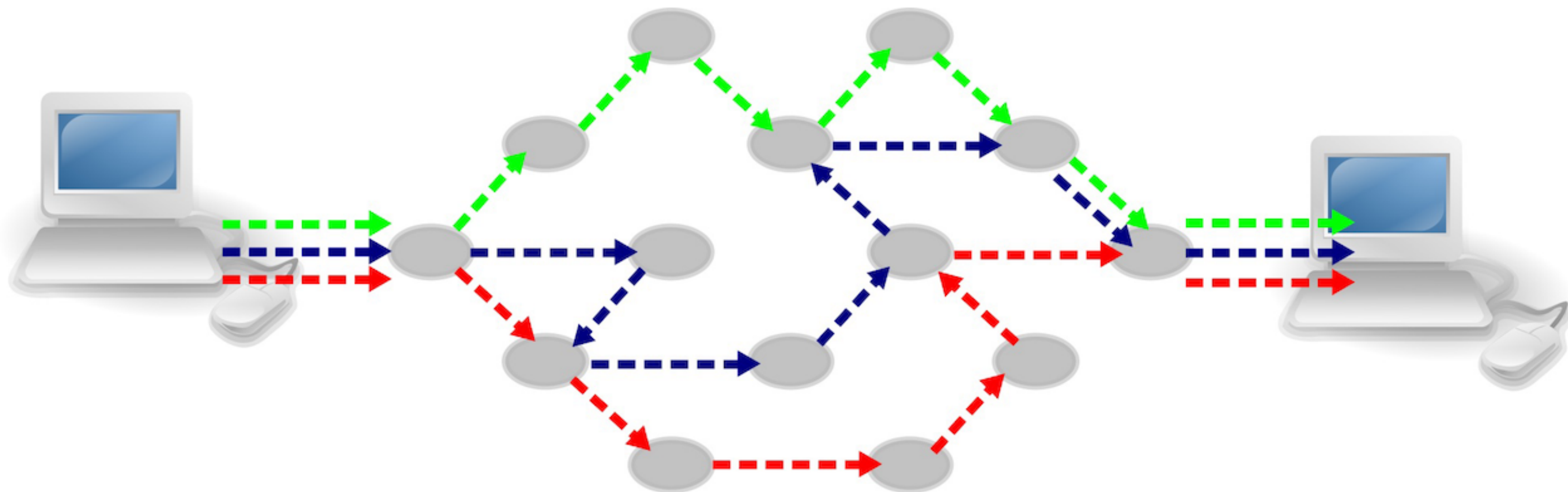


Packet switching and the nodes

- **Each communication node must know how to assemble/disassemble information to/from the packets**
- **Each communication node must know which link should be used to send a packet for the given destination**
- **Packets can be lost; relaying nodes cannot detect a lost packet**

Packet (dis)assembly issues

- **The sequence of delivered packets may differ from that of the sender intents; holding the out-of-sequence packets are required**
- **Retransmission is required to recover a lost packet for a reliable communication**

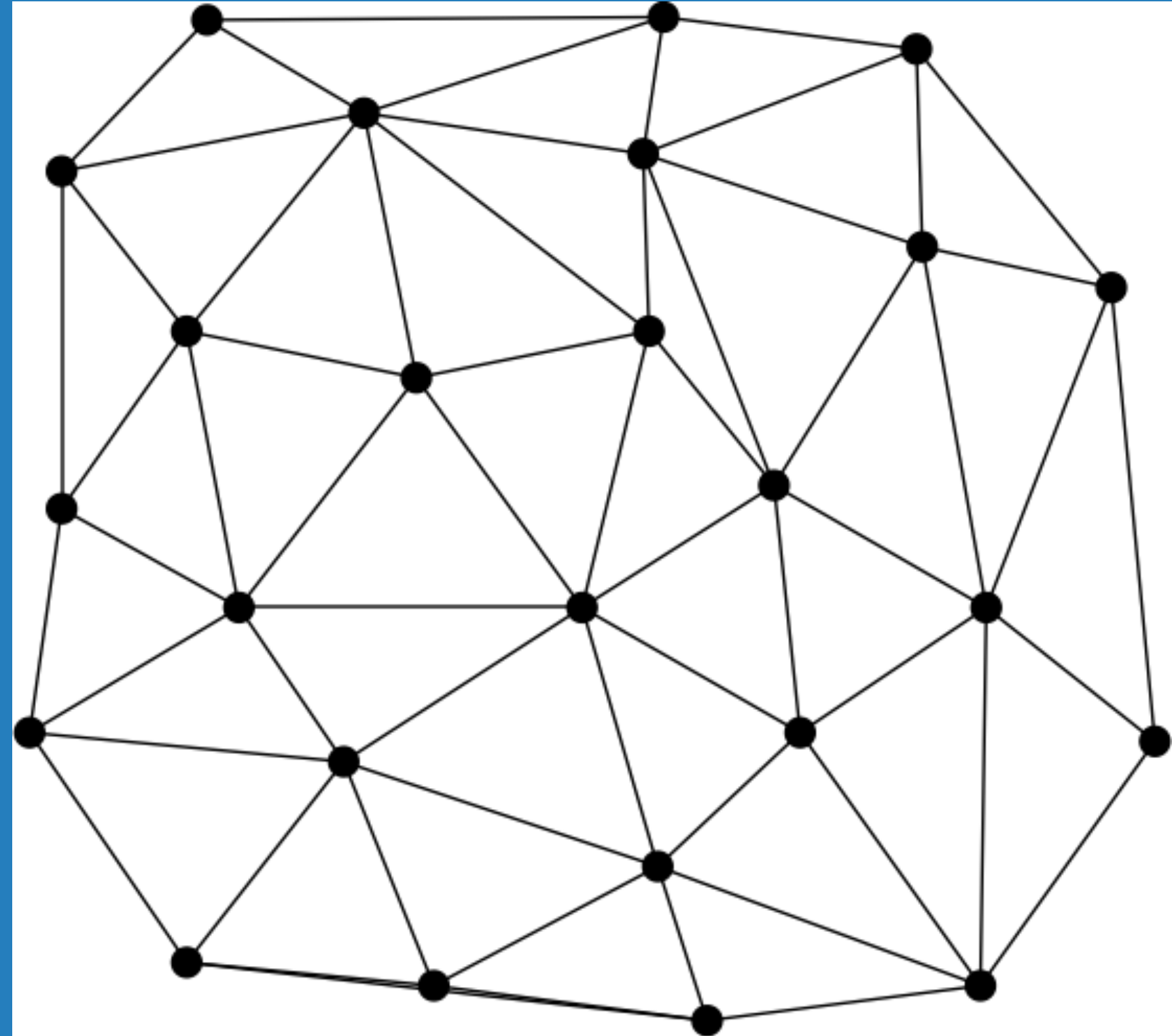


Packet switching enables

- Changing the packet relay routes *during* the communication
- Using multiple routes for a single communication link
- Aggregating multiple communication links into a physical link
- Connectionless *and* connection-oriented communication simultaneously

Truly distributed networks are feasible by packet switching

- No centralized nodes
- Each link can be utilized by all nodes
- A disconnection of the link will not be fatal so long as one link is connected to a node



Disadvantages of packet switching

- Each node must be able to form/generate and decode/interpret a packet
- Forming and decoding a packet takes time and the computing resources
- Reliability and latency can be a trade-off
- Relay nodes can be neutralized by denial-of-service attacks
- Difficult to manage

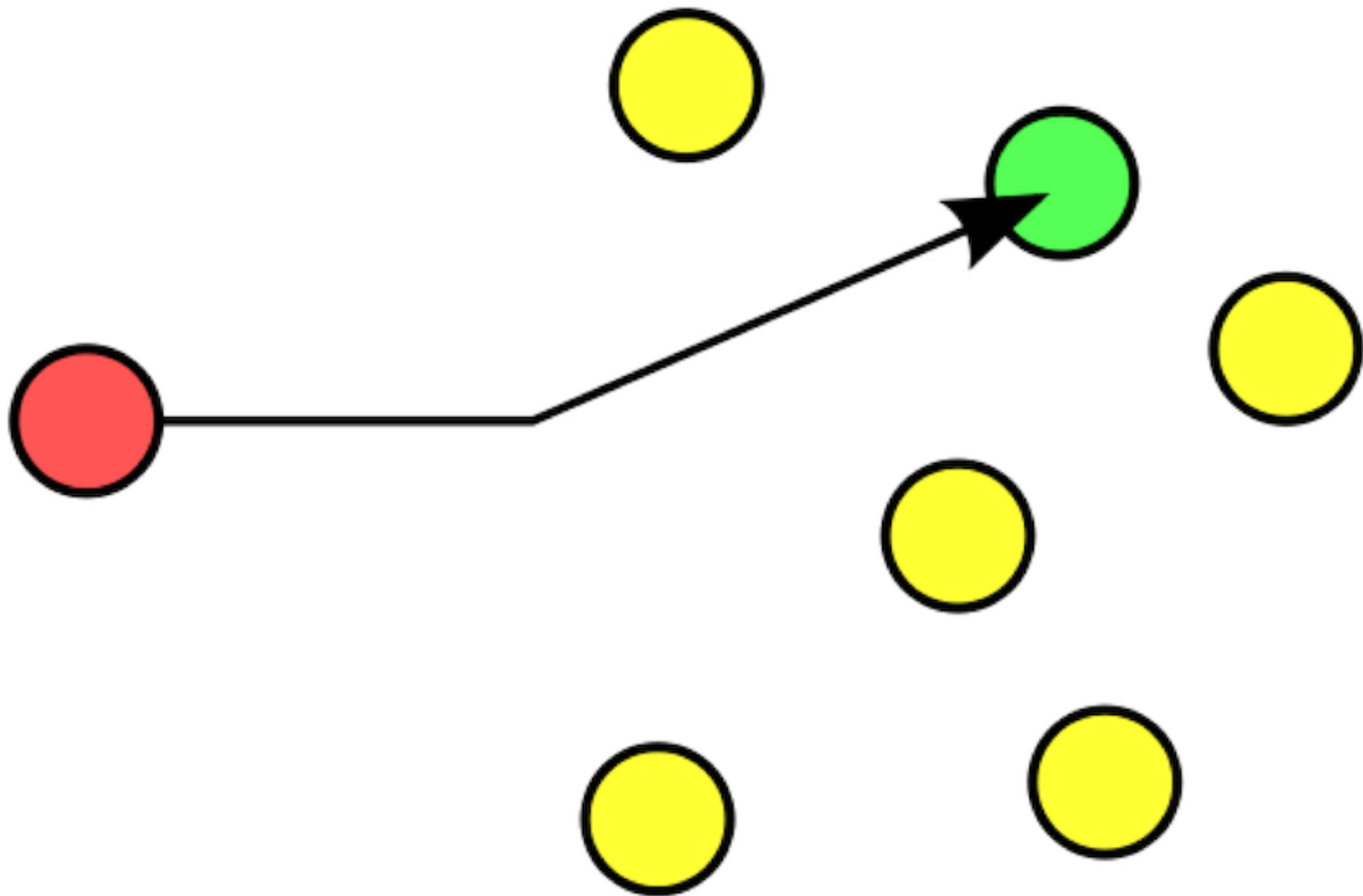
Routing basics

Various aspects of routing

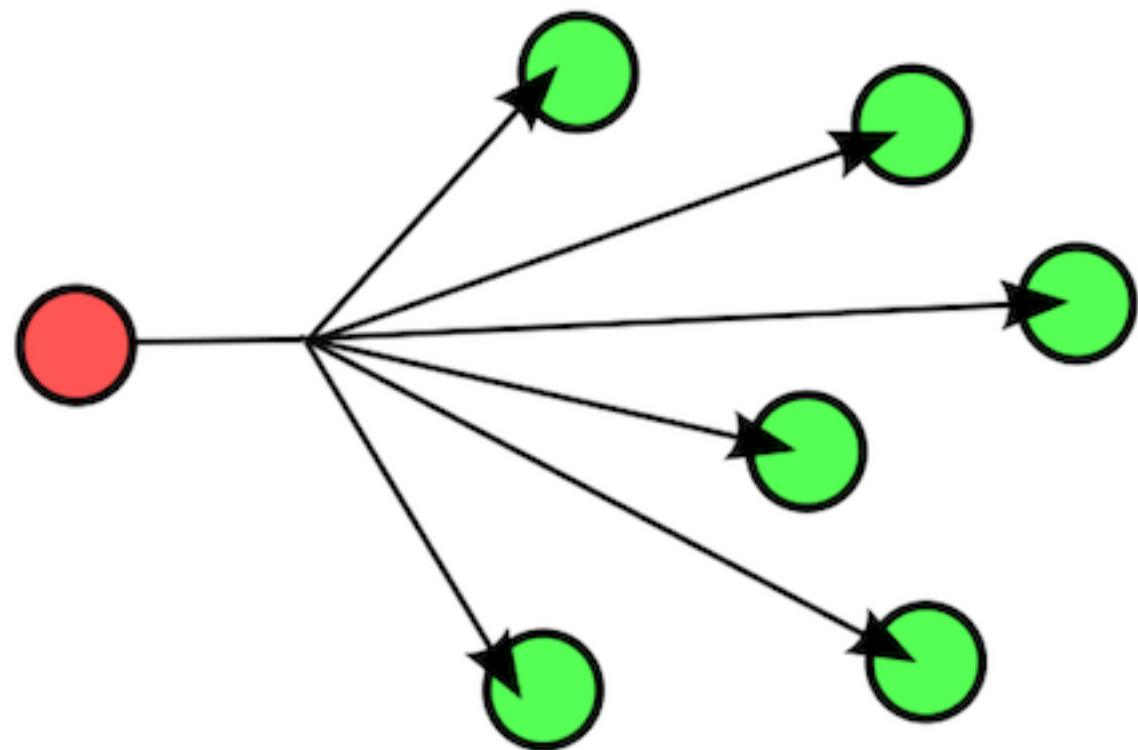
- **Delivery**
- **Addresses**
- **Static or dynamic**
- **Route aggregation**
- **Security**

Delivery schemes

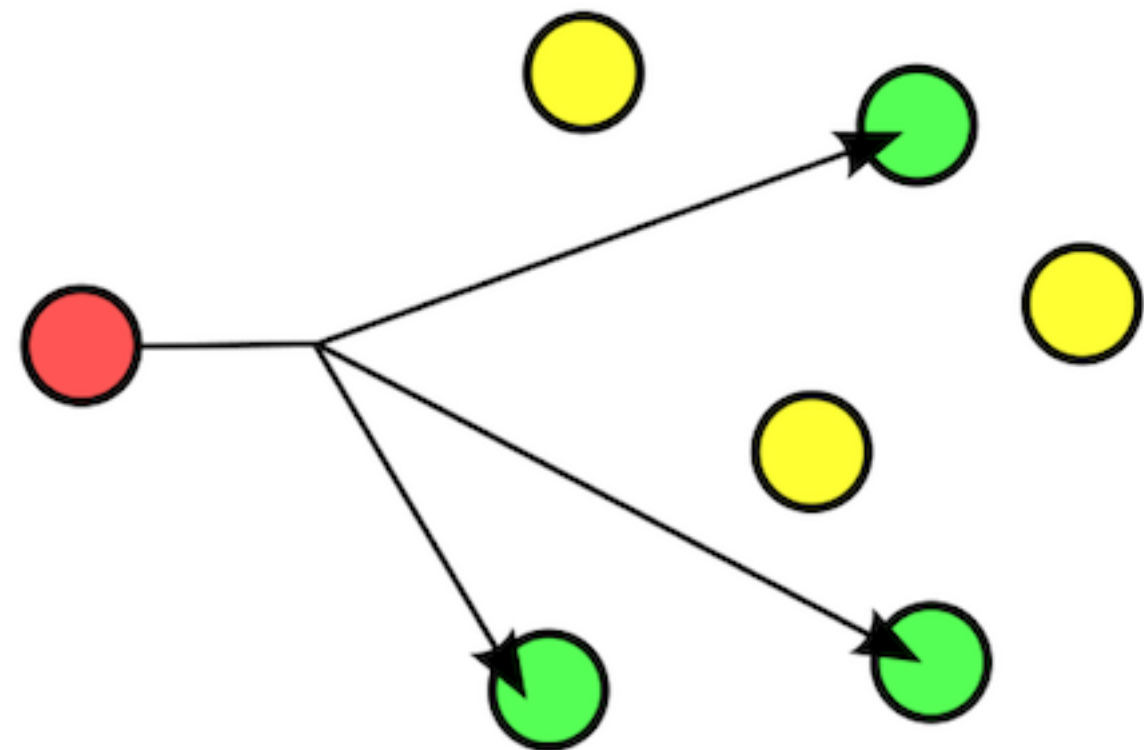
- **Unicast**
- **Broadcast/Multicast/Anycast**



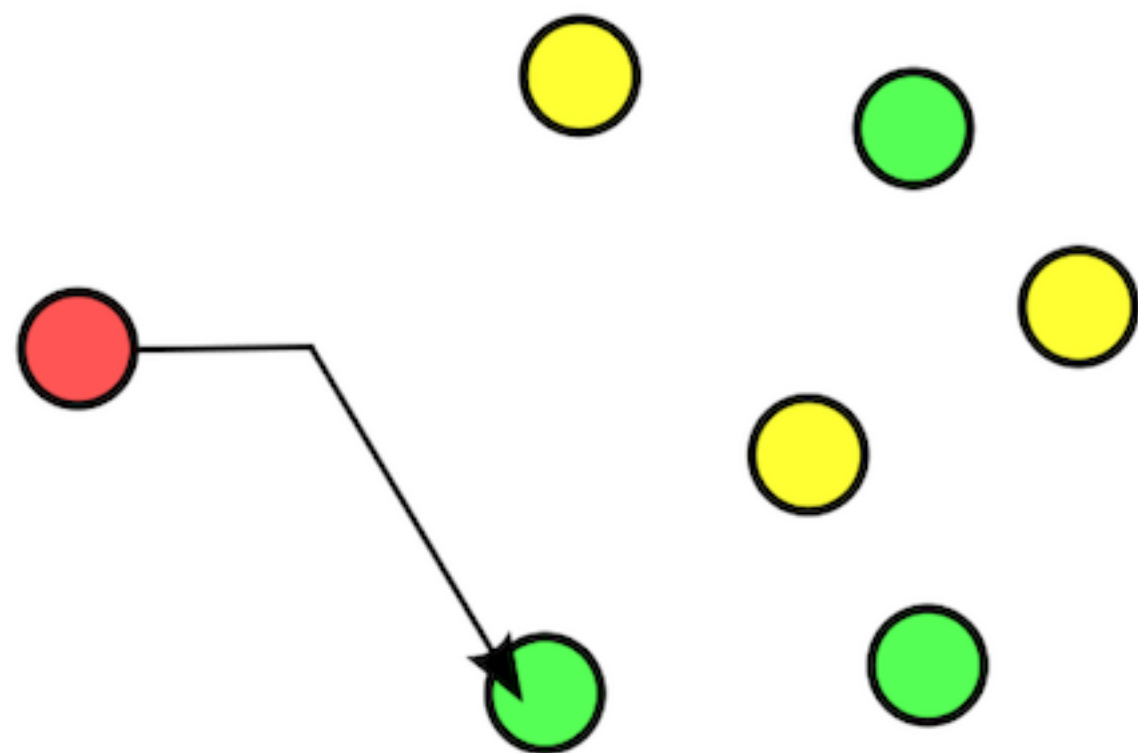
Unicast



Broadcast



Multicast



Anycast

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